

Real-Time Weather Data Pipeline

A comprehensive cloud-based solution for collecting, processing, and visualizing live weather data through integrated batch and streaming architectures.

Project Team

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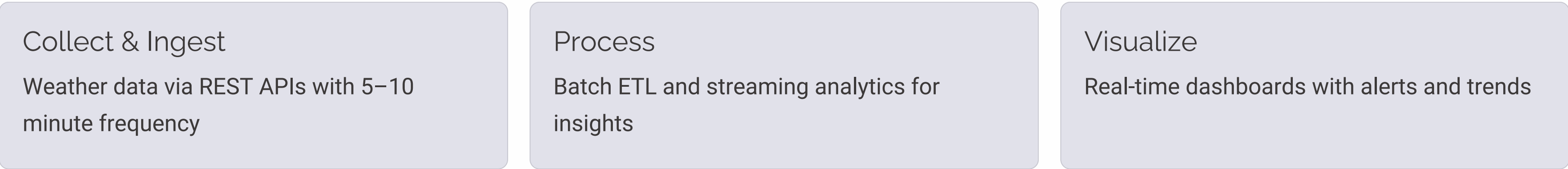
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Project Planning & Objectives

This project constructs a robust **real-time weather data pipeline** that seamlessly integrates data ingestion from weather APIs, processes information through both batch and streaming methodologies, and delivers actionable insights via interactive dashboards. The solution demonstrates enterprise-grade data engineering practices spanning the complete lifecycle from collection to visualization.



Project Milestones

Phase	Description	Tools	Week
1	Data Ingestion from Weather API	Python, Requests	1
2	Batch ETL Pipeline Implementation	Python	2
3	Streaming Pipeline & Alert System	Apache airflow	3
4	Dashboard & Reporting	Power BI	4

Database Design & Data Architecture

Data Source & Ingestion

The pipeline sources live weather data from industry-standard providers such as OpenWeatherMap or VisualCrossing , receiving JSON responses containing temperature, humidity, pressure, wind speed, and precise timestamps. Data collection occurs at regular 5–10 minute intervals, ensuring comprehensive temporal coverage for both real-time monitoring and historical analysis.

01

Ingestion Layer

Python scripts execute API calls to fetch raw weather data

02

Storage Layer

Raw JSON data persisted in SQL Database

03

Processing Layer

Batch ETL creates daily summaries; streaming detects anomalies

04

Visualization Layer

Power BI dashboards present metrics, trends, and alerts



UI/UX Dashboard Design

The dashboard represents the critical user-facing layer, delivering **real-time weather insights** through an intuitive, visually sophisticated interface. Design principles emphasize clarity, responsiveness, and accessibility while enabling rapid decision-making through live metrics and actionable alerts.

Key Design Principles

Simplicity

Minimalist design focusing exclusively on critical live metrics and eliminating visual clutter

Color Coding

Blue tones indicate normal conditions; red signals anomalies for instant pattern recognition

Responsiveness

Auto-refresh and live update capabilities ensure data freshness across all devices

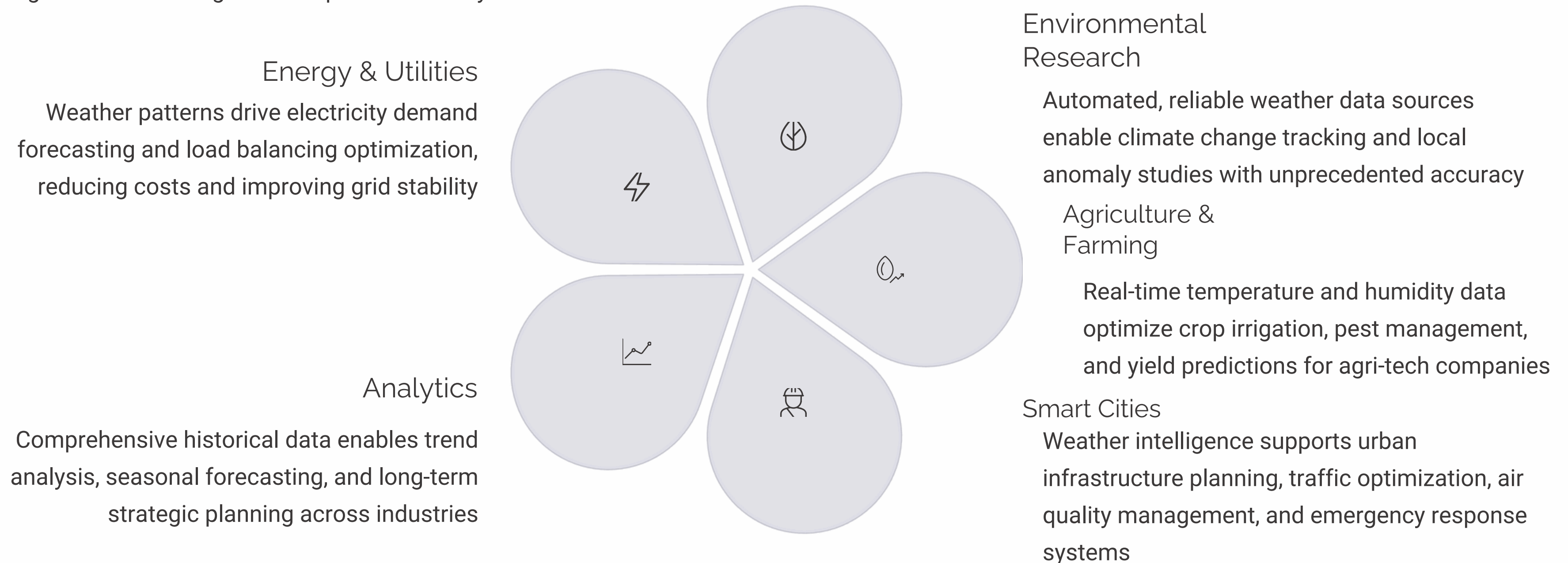
Dashboard Components

- **Weather conditions by city**
- **Average wind speed**
- **Average precipitation by city**
- **Average humidity by city**
- **Average temperature by city**



Stakeholder Impact & Business Value

This weather data pipeline delivers transformative value across multiple sectors, enabling data-driven decision-making and operational optimization through real-time intelligence and predictive analytics.



Strategic Advantage: Organizations leveraging this pipeline gain competitive intelligence through early anomaly detection, precise demand forecasting, and data-driven operational adjustments, translating to measurable cost savings and improved resilience.